Activity VII: Soft Biometric (Keystroke)

Given a team of two persons. Use the knowledge of digraph and trigraph (explained by your professor) to create a template for you and your teammate. Please measure (at least) time between 2-consecutive keys and total time. Measure whether your software can correctly identify the person.

digraph.py > ...

```
import keyboard
import time


def create_digraph():
    sumOfDigraph = {}
    countOfDigraph = {}
    prev = "enter"
    t = time.time_ns()
    while 1:
        key = keyboard.read_key()
        if key == "enter":
            break
        if prev != "enter":
            t = time.time_ns()
            if prev != "esc":
                if prev != key:
                    if (prev, key) in sumOfDigraph:
                        countOfDigraph[prev, key] += 1
                        sumOfDigraph[prev, key] += t - prev_time
                    else:
                        countOfDigraph[prev, key] = 1
                        sumOfDigraph[prev, key] = t - prev_time
        prev = key
        prev_time = t
    for p in sumOfDigraph:
        sumOfDigraph[p] = sumOfDigraph[p] / countOfDigraph[p]
    return sumOfDigraph


def validate(person1, person2):
    count = 0
    sum_error = 0
    for p in person1:
        if p in person2:
            sum_error = abs(person2[p] - person1[p])
            count += 1
    val = sum_error / count
    if val > 20_000_000:
        return False
    else:
        return True
```

```python
def main():
    print("create person1")
    person1 = create_digraph()
    print("\n",person1)
    print("\ntype 'esc' to create another")
    keyboard.wait("esc")
    print("create person2")
    person2 = create_digraph()
    print("\n",person2)
    while True:
        print("\ntype 'esc' to check who you are")
        keyboard.wait("esc")
        print('type "hello world my name is"')
        anonymous = create_digraph()
        check1 = validate(person1, anonymous)
        check2 = validate(person2, anonymous)
        if check1:
            print("\nYou are person1")
        elif check2:
            print("\nYou are person2")
        else:
            print("Who are you ?")


if __name__ == "__main__":
    main()
```

Please also answer the following questions.

```
jirawat@Macintosh ~ % sudo /usr/local/bin/python3 /Users/jirawat/Downloads/Activity_XI-Biometric/digraph.py
create person1
hello world my name is
 {(4, 14): 375803000.0, (14, 37): 285689000.0, (37, 31): 160777000.0, (31, 'space'): 101610000.0, ('space', 13): 269005000.0, (13, 31): 196837000.0, (31, 15): 298469000.0, (
15, 37): 214913000.0, (37, 2): 725918000.0, (2, 'space'): 183834000.0, ('space', 46): 506201000.0, (46, 16): 370199000.0, (16, 'space'): 464161000.0, ('space', 45): 20067200
0.0, (45, 46): 383703000.0, (46, 14): 140599000.0, (14, 'space'): 175760000.0, ('space', 34): 572579000.0, (34, 1): 527635000.0}

type 'esc' to create another

create person2
Hello World My Name is
 {('delete', 'right shift'): 528796000.0, ('right shift', 4): 78230000.0, (4, 'right shift'): 96185000.0, ('right shift', 14): 202014000.0, (14, 37): 133249000.0, (37, 31):
88339000.0, (31, 13): 57102000.0, (13, 'space'): 214634000.0, ('space', 'delete'): 246675000.0, ('delete', 'space'): 354352000.0, ('space', 'right shift'): 319203000.0, ('ri
ght shift', 13): 61743500.0, (13, 'right shift'): 67172000.0, (13, 31): 119678000.0, (31, 15): 96235000.0, (15, 37): 169856000.0, (37, 2): 45981000.0, (2, 'space'): 17154900
0.0, ('right shift', 46): 172622000.0, (46, 'right shift'): 3620000.0, ('right shift', 16): 1037410000.0, (16, 'space'): 366123000.0, ('right shift', 45): 87525000.0, (45, '
right shift'): 66687000.0, (46, 14): 76467000.0, (14, 'space'): 69790000.0, ('space', 34): 227123000.0, (34, 1): 459856000.0}

type 'esc' to check who you are

type "hello world my name is"
^[Hello World My Name is
You are person2
```

1. How many words do we need to correctly identify the person?
   - From the experiment, it succeeded to identify us on the first time which contained only 5 words, even if we had mistyped some words in different uppercase & lowercase due to miscommunication.

2. Do you think this method is scalable? (to thousand persons) for either recognition system or identification system. Please provide your analysis.
   - Theoretically, I think it should be scalable with more required words or sentences or up to a paragraph, identifying digraph validation of a spent time for typing between each pair of words (not only letters).
   - I mentioned “Theoretically”, because, in a real world, a user must be exhausting typing those validating phase as much as the number of users increases.

3. Will you use this kind of authentication in your system? Please also provide a reason.
   - No. I have stated the reason in the previous answer.